Consider the equilibrium that exists for a saturated aqueous solution of $PbCl_2$.

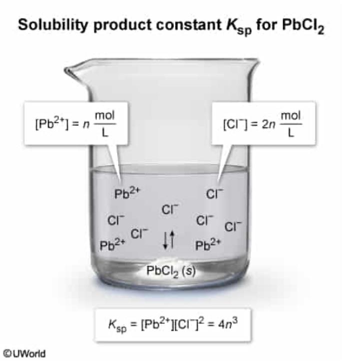
$$\mathrm{Pb} \qquad \mathrm{Cl}_2 \qquad (s) \qquad \rightleftarrows$$

Which expression gives the solubility product constant K_{sp} for $PbCl_2$ if the $[Pb^{2+}] = n$ mol/L?

- A. n^2 [12%]
- B. $2n^2$ [20%]
- C. n^3 [30%]
- ✔ D. $4n^3$ [36%]

Correct
36% answered correctly

Explanation:



For an ionic compound Z, which dissociates into ions X and Y when placed in water, an equilibrium will be established in which the rate of dissolution (forward reaction) equals the rate of precipitation (reverse reaction).

$$aZ_{(s)} \overset{K_{sp}}{\rightleftarrows} bX_{(aq)} + cY_{(aq)}$$

This equilibrium position is indicated by the solubility product constant K_{sp} which provides a measure of the solubility of a compound at a given temperature. The general K_{sp} expression is written as:

$$K_{sp} = [X]^b[Y]^c$$

where [X] and [Y] are the molar concentrations of the X and Y ions, and *b* and *c* are the respective stoichiometric coefficients of the **balanced dissolution reaction**. The K_{sp} is the product of the maximum possible concentrations of each ion in the solution under the given conditions. As a result, the K_{sp} represents the **limit of solubility** for the compound (a saturated solution).

Applying these concepts to the solubility equilibrium of $PbCl_2$, the K_{sp} for the compound is:

$$K_{sp} = [Pb^{2+}][Cl^-]^2$$

The equilibrium shows that 2 moles of Cl^- are present for every 1 mole of Pb^{2+}. As a result, if $[Pb^{2+}] = n$ mol/L, then $[Cl^-] = 2n$ mol/L:

$$\frac{n\ \cancel{\text{mol Pb}^{2+}}}{\text{L}} \times \frac{2\ \text{mol Cl}^-}{1\ \cancel{\text{mol Pb}^{2+}}} = \frac{2n\ \text{mol Cl}^-}{\text{L}}$$

Substituting these values into the K_{sp} expression for $PbCl_2$ gives:

$$K_{sp} = (n)(2n)^2 = (n)(4n^2) = 4n^3$$

(Choices A, B, and C) These expressions result from one or both of the following calculation errors: using $[Cl^-] = n$ mol/L, or failing to square the $[Cl^-]$ term.

Educational objective:
The solubility product constant K_{sp} represents the limit of solubility for a compound, and it is defined as the product of the molar concentrations of the dissolved species each raised to the power of their respective coefficients from the balanced dissolution reaction.

Time spent:0 sec
Subject:
General Chemistry

QID:402636
System:
Solutions and Electrochemistry

An ionic hydroxide compound formed from an unknown metal M exhibits the equilibrium

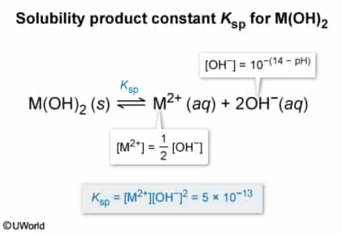
$$M(OH)_2(s) \rightleftarrows M^{2+}(aq) + 2\ OH^-(aq)$$

which results in a saturated aqueous solution wherein the molar concentration of $M^{2+}(aq)$ is half the molar concentration of $OH^-(aq)$. If the solution pH is 10, the solubility product constant K_{sp} of the compound is:

- A. 5.0×10^{-31}. [5%]
- B. 5.0×10^{-13}. [53%]
- C. 1.0×10^{-12}. [27%]
- D. 1.0×10^{-8}. [13%]

Correct
54% answered correctly

Explanation:



For an ionic compound Z that dissociates into ions X and Y when placed in water, an equilibrium will be established in which the rate of dissolution (forward reaction) equals the rate of precipitation (reverse reaction).

$$a\ \mathrm{Z}(s) \overset{K_{sp}}{\rightleftarrows} b\ \mathrm{X}(a$$

This equilibrium position is indicated by the **solubility product constant K_{sp}**, which provides a measure of the solubility of a compound at a given temperature. The general K_{sp} expression is written as:

$$K_{sp} = [\mathrm{X}]^b [\mathrm{Y}]^c$$

where [X] and [Y] are the molar concentrations of the X and Y ions, and *b* and *c* are the respective coefficients of the **balanced dissolution reaction**. The K_{sp} is the product of the maximum possible concentrations of each ion in the solution under the given conditions.

Applying these concepts to the solubility equilibrium of $M(OH)_2$, the K_{sp} expression for the compound is:

$$K_{sp} = [\mathrm{M}^{2+}][\mathrm{OH}$$

Using the pH of the saturated solution, the $[OH^-]$ can be determined by first evaluating the pOH.

$$\mathbf{pH} + \mathbf{pOH} = 14$$
$$\mathrm{pOH} = 14 - 10 = 4$$

The $[OH^-]$ is then determined from its relationship to pOH.

$$\mathrm{pOH} = -\mathbf{log}[$$
$$[\mathrm{OH}^-] = 10^{-\mathrm{pOH}}$$

The reaction coefficients are not included in the calculations with pH and pOH because the pH and pOH values directly measure the final *result* of the reaction and already account for the molar proportions of the reaction.

The $[M^{2+}]$ was not directly measured, but the mole ratio of the equilibrium reaction equation shows that 2 moles of OH^- are produced for every 1 mole of M^{2+}. As a result, $[M^{2+}]$ must be equal to half of $[OH^-]$.

$$1 \times 10^{-4} \frac{\cancel{\mathrm{mol\ OH^-}}}{\mathrm{L}} \times \frac{1\ \mathrm{mol\ M^{2+}}}{2\ \cancel{\mathrm{mol\ OH^-}}} = 0.5 \times 10^{-4} \frac{\mathrm{mol\ M^{2+}}}{\mathrm{L}}$$

$$[\mathrm{M}^{2+}] = 0.5 \times 10^{-4} = 5 \times 10^{-5} \frac{\mathrm{mol\ M^{2+}}}{\mathrm{L}}$$

Substituting the values of $[M^{2+}]$ and $[OH^-]$ into the K_{sp} expression and applying the rules for math operations with scientific notation gives:

$$K_{sp} = [\mathrm{M}^{2+}][\mathrm{OH}^-]^2 = (5 \times 10^{-5})(1 \times 10^{-4})^2 = (5 \times 10^{-5})(1 \times 10^{-8}) = 5 \times 10^{-13}$$

(Choice A) A value of 5.0×10^{-31} is obtained if 10^{-pH} is used for $[OH^-]$ instead of 10^{-pOH}.

(Choice C) A value of 1.0×10^{-12} results from assuming that $[M^{2+}] = [OH^-]$.

(Choice D) A value of 1.0×10^{-8} is calculated by assuming that $[M^{2+}] = [OH^-]$ and also failing to square the $[OH^-]$ term.

Educational objective:
The solubility product constant K_{sp} is defined as the product of the maximum molar concentrations of the dissolved species, each raised to the power of their respective coefficients from the balanced dissolution reaction. An equilibrium involving $[OH^-]$ can be evaluated from the pH given that pH + pOH = 14 and pOH = $-\log[OH^-]$.

Time spent:0 sec
Subject:
General Chemistry

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System:
Solutions and Electrochemistry